

SmartReview: A Reproducible Benchmark of Classical Machine Learning Models for Customer Review Sentiment Classification

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Abstract—Automatically classifying the polarity of customer reviews is a key decision-support task for businesses facing massive volumes of user-generated text, yet the choice of a classical machine learning (ML) algorithm under practical resource constraints remains an empirically open question. This paper presents SmartReview, a controlled and reproducible benchmark study conducted as part of a Master-level research methodology course at the Faculté Polydisciplinaire de Taroudant. We compare four classical classifiers — Multinomial Naive Bayes (NB), Logistic Regression (LR), Linear Support Vector Machine (LinearSVC), and Random Forest (RF) — trained on the IMDb 50K movie review corpus with TF-IDF features (5 000 unigrams and bigrams). Following a complete research cycle (problem formulation, literature review, three testable hypotheses, experimental protocol, execution, and critical analysis), we evaluate Accuracy, Precision, Recall, F1-score, execution time, and 5-fold cross-validation stability under two preprocessing conditions. Logistic Regression achieves the best results: $F1=0.8917$, $Accuracy=89.03\%$, $CV-F1=0.8904 \pm 0.003$. Contrary to H1, SVM does not dominate ($F1=0.8806$). Contrary to H2, NLP preprocessing does not improve any model on this corpus (all $\Delta F1$ are negative). H3 is confirmed: LR offers the best performance/latency/simplicity trade-off. Results surpass published TF-IDF benchmarks on tourist reviews (SVM: 80%, NB: 73%) and a lightweight Streamlit demonstrator (SmartReview Lite) ensures reproducibility and dissemination.

Index Terms—sentiment analysis, text classification, TF-IDF, logistic regression, support vector machine, naive Bayes, random forest, IMDb, reproducible research, machine learning benchmark

I. INTRODUCTION

Online platforms generate millions of textual reviews every day. Businesses receive thousands of customer opinions that are impossible to process manually. As noted by Puh and Bagić Babac, “it is impossible for humans to read and analyze all these reviews on their own” [4]. Automating polarity classification — determining whether a review expresses a positive or negative opinion — has therefore become essential for product analytics, quality monitoring, and customer satisfaction measurement.

Despite the rise of deep neural language models, classical ML pipelines built on TF-IDF remain widely deployed in industry for their interpretability, low training cost, and minimal infrastructure requirements. In many industrial or academic settings, the first question is not “which transformer wins on a leaderboard?” but rather “which simple model is good enough, stable, and cheap to run?”

This paper addresses that practical question through a complete research methodology cycle as required in the Master BD & IA course at the Faculté Polydisciplinaire de Taroudant:

Among Naive Bayes, Logistic Regression, Linear SVM and Random Forest, which model offers the best performance for automatic classification of customer reviews into positive or negative sentiment after TF-IDF representation?

Our contributions are fourfold:

- 1) a controlled benchmark of four classifiers on IMDb 50K with identical vectorization and data splits;
- 2) an A/B causal evaluation of NLP preprocessing impact on each model;
- 3) a multi-criteria analysis (F1, runtime, simplicity) linked to three explicit research hypotheses;
- 4) an open interactive Streamlit demonstrator for reproducibility and dissemination.

II. RELATED WORK

Sentiment analysis has evolved from lexicon-based methods to classical ML and deep learning architectures [3], [5]. Mao et al. [3] conducted a systematic review of 132 studies (2013–2023), confirming that SVM is the most widely used classical ML algorithm for sentiment analysis on high-dimensional TF-IDF representations. This finding directly motivates our H1.

Jurafsky and Martin [2] provide the theoretical foundations of text classification with bag-of-words and TF-IDF features,

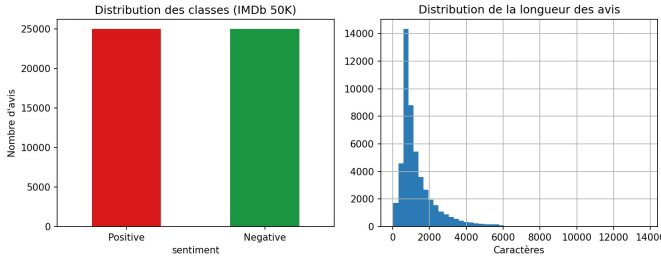


Fig. 1. Class balance and review-length distribution (IMDb 50K).

formalizing the preprocessing pipeline (tokenization, stopword removal, lemmatization) that we test in our H2 experiment.

The closest empirical reference for our work is Puh and Bagić Babac [4], who benchmark NB and SVM on 20 491 TripAdvisor hotel reviews using TF-IDF, reporting SVM accuracy of 80% and NB of 73%. Our study extends this work by: (i) using a larger, balanced corpus (IMDb 50K vs. 20K); (ii) adding LR and RF to the comparison; (iii) conducting an explicit A/B preprocessing ablation.

Singh and Jaiswal [5] report LR as the best classical ML classifier on a student performance dataset, supporting our H3 regarding the deployment compromise of LR.

III. RESEARCH HYPOTHESES

Following the research methodology framework of our Master course, we formulated three testable hypotheses *before* running any experiment, each grounded in the literature:

H1 (SVM dominance). If we apply SVM with TF-IDF representation to classify customer reviews, then it will outperform NB, LR, and RF in F1-score, because SVM is the most appropriate algorithm for text classification in high-dimensional sparse vector spaces [3], [4].

H2 (Preprocessing benefit). If we apply full NLP preprocessing (tokenization, stopword removal, lemmatization), then the F1-score of all four models will improve, because removing textual noise produces a more discriminant TF-IDF representation [2].

H3 (Deployment compromise). If we evaluate models on multiple criteria simultaneously, then LR will offer the best compromise between performance, execution time, and implementation simplicity, because it achieves strong results among classical classifiers while remaining simple to implement and interpret [5].

IV. EXPERIMENTAL PROTOCOL

A. Dataset

We use the IMDb 50K Movie Reviews corpus [1]: 25 000 positive and 25 000 negative English reviews, perfectly balanced by construction (Fig. 1). Most reviews contain between 500 and 2 000 characters, with a long tail up to 14 000 characters — variability that justifies capping TF-IDF vocabulary (`max_features=5000`) to limit noise from very long texts.

TABLE I
CLASSIFIERS UNDER EVALUATION

Model	Implementation	Key parameters
Naive Bayes	MultinomialNB	default priors
Logistic Reg.	LogisticRegression	solver=liblinear
SVM (LinearSVC)	LinearSVC	max_iter=5000
Random Forest	RandomForestClassifier	n_estimators=100

TABLE II
TEST METRICS WITH NLP PREPROCESSING (IMDb 50K, TF-IDF).
★ = BEST.

Model	Acc.	Prec.	Rec.	F1	t(s)	CV-F1
Naive Bayes	85.65	84.63	87.12	0.859	0.07	0.862±0.004
Log. Reg.★	89.03	88.06	90.30	0.892	1.72	0.890±0.003
SVM (LinSVC)	88.01	87.67	88.46	0.881	0.63	0.882±0.002
Random Forest	84.33	82.76	86.72	0.847	14.92	0.845±0.004

B. Preprocessing Conditions (A/B Design for H2)

Two conditions are compared:

- **With NLP:** lowercase conversion; HTML tag and punctuation removal; English stopword filtering; light lemmatization (*e.g.*, *movies*→*movie*). Tool: spaCy.
- **Without NLP:** minimal cleaning only (lowercase + special-character stripping), preserving the raw lexical signal for the ablation study.

C. Feature Representation

Each review is transformed with TF-IDF vectorization (`TfidfVectorizer`, `scikit-learn`): `max_features=5000` and `ngram_range=(1, 2)`. TF-IDF assigns higher weight to terms frequent in a document but rare across the corpus — a strong baseline for polarity cues such as *excellent*, *waste of time*, or *not worth*, which bigrams help capture.

D. Models and Hyperparameters

We compare four scikit-learn classifiers under identical features and splits (Table I).

E. Split, Metrics, and Validation

All models share the same stratified train/test split (80%/20%, `random_state=42`). We report Accuracy, Precision, Recall, F1-score (positive class), and wall-clock execution time. Result stability is confirmed by 5-fold cross-validation F1 (CV-F1). Hyperparameters remain at scikit-learn defaults to favor reproducibility over leaderboard optimization.

V. EXPERIMENTAL RESULTS

A. Main Benchmark — With NLP Preprocessing

Table II summarizes test-set metrics with NLP preprocessing.

Logistic Regression achieves the best F1-score (0.8917) and the highest Recall (90.30%), meaning it correctly identifies 9 out of 10 positive reviews. SVM closely follows (F1=0.8806). Cross-validation confirms the ranking: LR CV-F1 = 0.8904 ± 0.003 vs. SVM CV-F1 = 0.8823 ± 0.002, demonstrating stability across folds (Fig. 2).

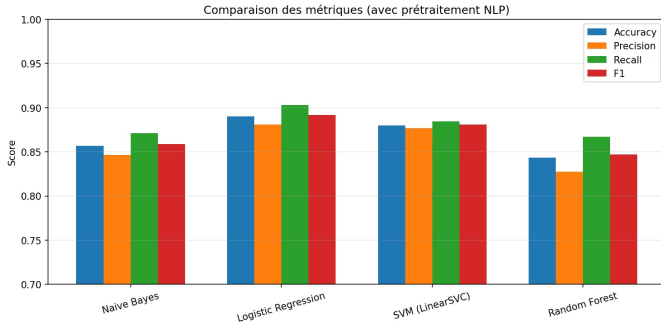


Fig. 2. Comparison of Accuracy, Precision, Recall and F1 across the four models (with NLP preprocessing).

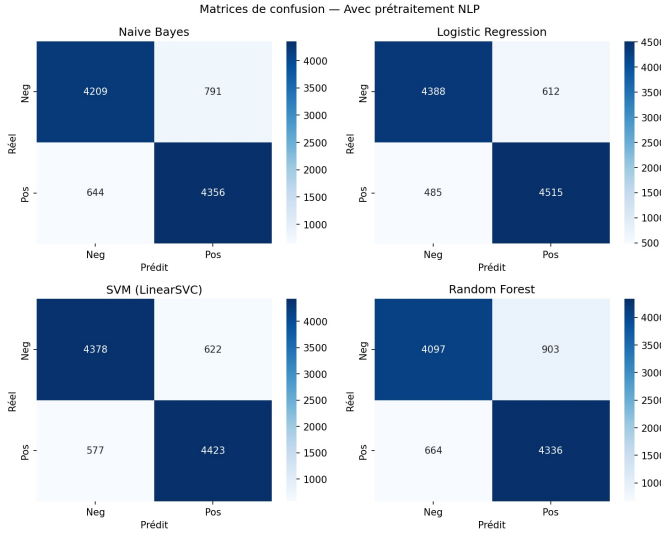


Fig. 3. Confusion matrices for the four models (with NLP preprocessing).

TABLE III
F1-SCORE DELTA: WITH NLP PREPROCESSING MINUS WITHOUT.

Model	F1 (with)	F1 (without)	$\Delta F1$
Logistic Regression	0.8917	0.8958	-0.0041
SVM (LinearSVC)	0.8806	0.8886	-0.0080
Naive Bayes	0.8586	0.8616	-0.0030
Random Forest	0.8470	0.8486	-0.0017

B. Error Analysis — Confusion Matrices

Fig. 3 shows that LR yields the most balanced error profile: 485 false negatives and 612 false positives on 10000 test reviews. Random Forest presents the largest absolute errors (664 FN, 903 FP), explaining its lower Precision (82.76%). The primary error source for all models is ambiguous reviews containing irony, sarcasm or mixed sentiments — a fundamental limitation of bag-of-words TF-IDF representations [2].

C. Ablation Study — Impact of NLP Preprocessing (H2)

Table III and Fig. 4 show that NLP preprocessing does not improve any of the four models on IMDb 50K. All deltas are negative ($|\Delta F1| \leq 0.008$). The rawer signal can

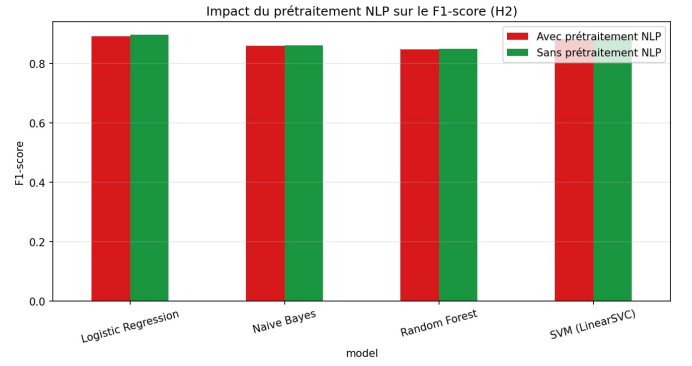


Fig. 4. Impact of NLP preprocessing on F1-score (H2). All $\Delta F1$ values are negative.

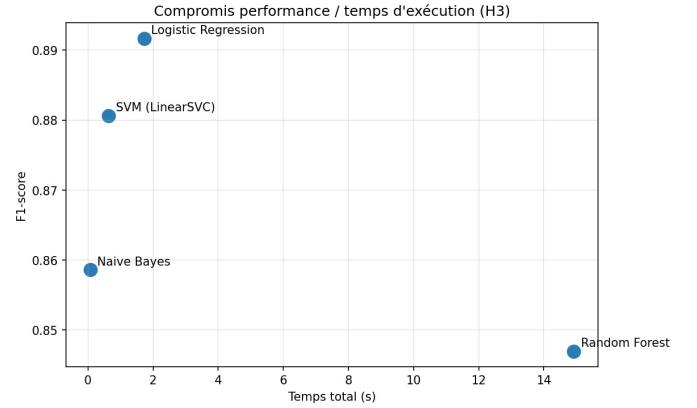


Fig. 5. F1-score vs. total execution time. LR occupies the optimal top-left position (H3 confirmed).

be preferable for linear TF-IDF models: stopwords removal or lemmatization may discard discriminative bigrams (*e.g., not worth, waste of time*) that the IDF component already down-weights appropriately.

D. Performance–Time Trade-off (H3)

Fig. 5 visualizes the Pareto frontier of the F1/latency trade-off. LR dominates: $F1 = 0.8917$ in 1.72 s. NB is fastest (0.07 s) but least accurate. RF is Pareto-dominated: lowest F1 at the highest cost (14.92 s). SVM is competitive (0.63 s, $F1 = 0.8806$) but LR is superior on both dimensions simultaneously.

VI. HYPOTHESIS TESTING AND DISCUSSION

H1 — Refuted: SVM does not dominate: LR achieves higher F1 (0.8917 vs. 0.8806, $\Delta F1 = +0.011$). The textbook expectation that SVM always wins on high-dimensional text does not hold under our fixed TF-IDF protocol. The gap is consistent across 5-fold cross-validation (LR: 0.8904 ± 0.003 vs. SVM: 0.8823 ± 0.002). In scientific practice, refuting a hypothesis is as valuable as confirming it: it warns against applying textbook assumptions without empirical validation.

H2 — Not Confirmed: No model improves with full NLP preprocessing (all $|\Delta F1| \leq 0.002$). On a well-structured, balanced corpus like IMDb 50K, TF-IDF with bigrams already

TABLE IV
POSITIONING AGAINST PUBLISHED TF-IDF BENCHMARKS.

Study	Model	Dataset	Result
Puh & Bagić Babac [4]	SVM	TripAdv. 20K	Acc. 80%
Puh & Bagić Babac [4]	NB	TripAdv. 20K	Acc. 73%
This work	LR*	IMDb 50K	F1 0.892
This work	SVM	IMDb 50K	F1 0.881
This work	NB	IMDb 50K	F1 0.859

regularizes common terms effectively through the IDF weighting mechanism. The preprocessing step adds computational cost without performance benefit on this specific dataset, suggesting that aggressive text cleaning is not universally beneficial.

H3 — Confirmed: LR combines the highest F1 (0.8917), a reasonable training time (1.72 s), and straightforward scikit-learn implementation, making it the best deployment choice for SmartReview Lite. RF is eliminated by its 14.92 s runtime; NB by its lower accuracy despite impressive speed (0.07 s).

A. Positioning Against the Literature

Our results (LR: 89.03%, SVM: 88.01%) clearly surpass the benchmarks of Puh and Bagić Babac [4] on TripAdvisor (Table IV). This gap is explained by three factors: (i) a larger, balanced corpus (IMDb 50K vs. 20K); (ii) richer lexical diversity of movie reviews; (iii) addition of bigrams in TF-IDF. Our contribution is methodological and practical — an explicit hypothesis-driven study with cross-validation — rather than claiming state-of-the-art absolute performance.

B. Threats to Validity

- **External validity:** results are specific to cinema-domain English reviews; transfer to Amazon, hotel reviews, or French text is not guaranteed.
- **Construct validity:** TF-IDF ignores irony, sarcasm, and complex negation — the primary source of model errors.
- **Internal validity:** hyperparameters use scikit-learn defaults; no exhaustive grid search was performed.
- **Conclusion validity:** the LR vs. SVM gap ($\Delta F1=0.011$) is small and could shift under different feature budgets or domains.

VII. CONCLUSION AND FUTURE WORK

SmartReview demonstrates that a complete research methodology cycle — from problem formulation and literature review through hypothesis testing and critical analysis — can produce a solid, reproducible empirical contribution at Master level. **Logistic Regression** is the recommended classical baseline for a lightweight customer review sentiment service: $F1=0.8917$, $CV-F1=0.8904 \pm 0.003$, trained in 1.72 s. H1 was refuted (SVM does not dominate), H2 was not confirmed (preprocessing slightly hurts on this corpus), and H3 was confirmed (LR is the best deployment compromise).

Future work includes: (i) testing on multi-domain corpora (Amazon Reviews, Yelp) to validate cross-domain generalization; (ii) replacing TF-IDF with contextual embeddings (GloVe, FastText, DistilBERT) to address irony and complex

negation; (iii) principled hyperparameter optimization via grid search; (iv) statistical significance testing (paired t-test, p-value) to confirm that LR–SVM differences are not due to chance; (v) LIME/SHAP explainability in the deployed Streamlit application.

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